

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

KALAHSTHI RAMYA
192311149RA

CO3-AT1-Assignment

[← Back](#)

QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

Q5

Manufacturing Plant –
Producer-Consumer Problem
10 marks

5/5 answered

Q2

Hospital Patient Registration – User-Defined Exception

10
marks

Develop a Java program for a hospital patient registration system. The program accepts the patient's name, age, and body temperature.

Create suitable user-defined exception classes for invalid age and invalid temperature. Validate the input before registration. Use try-catch-finally and display a registration completion message from finally.

Sample Test Cases:

Input / Scenario Expected Result Purpose

Arun, 35, 98.6 Patient registered successfully Normal case

Meena, 12, 98.4 InvalidAgeException Boundary / exception case

Ravi, 45, 105.2 InvalidTemperatureException Invalid input

Kumar, -2, 98.1 InvalidAgeException Invalid input

Your Code

```
1 import java.util.*;  
2  
3 public class Main{  
4     public static void registerPatient(Strin
```

```
5      throws InvalidAgeException, InvalidTe
6
7      if(age < 0 || age > 130){
8          throw new InvalidAgeException("I
9      }
10     if(temperature < 95.0 || temperature
11         throw new InvalidTemperatureExce
12     }
13
14     System.out.println("Patient register
15 }
16
17 public static void main(String[] args){
18     Scanner sc = new Scanner(System.in);
19
20     String name = "";
21     int age = 0;
22     double temperature = 0.0;
23
24     if(sc.hasNext()){
25         name = sc.next();
26     }
27     if(sc.hasNextInt()){
28         age = sc.nextInt();
29     }
30     if(sc.hasNextDouble()){
31         temperature = sc.nextDouble();
32     }
33
34     try{
35         registerPatient(name, age, tempera
36     }catch(InvalidAgeException | Invalid
37         System.out.println(e.getMessage(
38     }finally{
39         System.out.println("registration
40         sc.close();
41     }
42 }
43 }
44
45 class InvalidAgeException extends Exception{
46     public InvalidAgeException(String msg){
47         super(msg);
48     }
49 }
```

```
50
51 class InvalidTemperatureException extends Ex
52     public InvalidTemperatureException(Strin
53         super(msg);
54     }
55 }
```



Input

Kumar
-2
98.1



Output

InvalidAgeException Invalid input
registration completion message from finally

✓ Submitted